

Using Assistive Technology to Generate Social Skills Use for Students With Emotional Behavior Disorders

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Abstract

Researchers have demonstrated the positive outcomes of using video self-modeling (VSM) with students with emotional behavior disorders (EBDs). VSM typically refers to teacher-created videos demonstrating desired skills or behaviors where the student is the video model performing the skill at a level higher than his or her typical display. The purpose of this article is to demonstrate the ease and effectiveness of a student-created VSM to generate social skills use in rural general education classrooms. Included are the steps three students with EBD used to produce VSM using an e-book application as an assistive technology to promote their performance of social skills in the general education environment. Suggestions are provided for school personnel who want to implement this assistive technology strategy in the general education setting.

Keywords

video self-modeling modeling, emotional behavior disorders, rural special education

Mr. Peters and Ms. Johnson were discouraged about Marcus, Claire, and Darren's engagement during classroom activities and their completion of assignments. All three students were identified as having an emotional behavior disorder (EBD) and received special education services to support them in the general education classroom. Despite their ability to do the work required of them and get along with peers and adults, they often did not attend to their schoolwork, disturbed other students, and failed to respond to their teacher's efforts to increase their enthusiasm for schoolwork. Claire was a frequent visitor to in-school suspension, and Marcus and Darren each had received two out-of-school suspensions during the last school year.

The teachers in this rural high school felt a shared responsibility for the success of all students in and outside of the classroom environment, but these three students were losing champions they had on the faculty. Their behavior was increasingly becoming a concern for school personnel who wanted to resolve the situations in house rather than refer them for outside of school assistance.

General Education Classroom Social Skill Instruction for Students With EBD

Interpersonal skills such as student use of active listening, effective communication, and recognizing/understanding others' point of view are essential social competencies for the development of healthy relationships and successful

integration into educational settings (Farrington et al., 2012). Students with an EBD often come to teachers' attention and are typically referred to special education because of deficits in these critical skills (Gresham, Cook, Crews, & Kern, 2004). The failure to acquire these skills often results in students engaging in aberrant behavior that affects relationships across all domains. Furthermore, it may impede the inclusion of students in school, thereby, creating a range of complex challenges in the provision of academic and behavioral supports (Lane, Barton-Arwood, Nelson, & Wehby, 2008; Shea & Bauer, 2012).

Gresham (2015) suggested the lack of social behaviors function as academic disablers, disrupting student social skill performances and, in turn, contributing to low academic achievement. These damaging interactions produce negative in-school events (e.g., failed courses, suspension, expulsion, school dropout) and dismal postsecondary outcomes including arrest and time spent in jail or on parole (Gage, Lierheimer, & Goran, 2012; Wagner, Newman, Cameto, & Levine, 2005). Students who exhibit excessive levels of behavioral problems are often sent to the office or placed in more restrictive environments (McLeskey, Landers, Williamson, & Hoppey, 2012). These types of exclusion from the general education

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classroom and the teacher's expertise in content decrease opportunities for practicing needed academic and social skills (Sugai, Sprague, Horner, & Walker, 2000).

Wentzel (2009) and Gresham (2015) purported that students with EBD who develop positive social relationships with peers, use self-regulation skills, and have higher rates of cooperation with others show higher academic engagement and achievement than those who do not. Other researchers have demonstrated that school personnel who implement guided socialization opportunities in inclusive school environments produce greater positive student outcomes related to interacting with peers and adults (Anderson, Moore, Godfrey, & Fletcher-Flinn, 2004; Rotheram-Fuller, Kasari, Chamberlain, & Locke, 2010). Muscott and Gifford (1994) noted that educators can maximize the effects of social skills instruction by providing students with the opportunity to observe and practice targeted skills. Bandura (1977) suggested that observational learning may not alone create behavior change, but that students must attend to modeled behavior and then be reinforced for performing that behavior. Furthermore, the student must find the skills they observe to be personally relevant and rewarding. Simply placing students with EBD in inclusive settings does not guarantee that improved social skills or academic achievement success will occur (Gutierrez, Hale, Gossens-Archuleta, & Sobrino-Sanchez, 2007; Scattone, 2007). Fortunately, general education classrooms can provide a rich environment for educators to teach, reinforce, and reward the performance of appropriate social skills by students with EBD when additional supports are incorporated (Johns, Crowley, & Guetzloe, 2005). One additional support may involve the use of assistive technology by general and special education teachers to facilitate the use of social skills.

Rural Culture and Students With EBD

Historically, students with EBD are most often placed in segregated settings due to problem behaviors that impede their learning and the learning of others (Alisauskas & Simkiene, 2013; Reid, Gonzalez, Nordness, Trout, & Epstein, 2004); however, data suggest that levels of inclusion are typically higher for those in rural settings (IDEA Data Center, 2014). Rural communities often hold different cultural values than urban communities, which affect their members' perspectives on the inclusion of individuals with disabilities (Chipp, Johnson, Brems, Warner, & Roberts, 2008). There is often a desire to include special education students due to strong community ties and an interdependence on one another to provide informal supports (Chipp et al., 2008; Nelson, Pomerantz, & Howard, 2007). However, limited resources and personnel make it necessary for all school staff to take part in the challenging work of educating and including students with EBD across their

school life span (Redding & Walberg, 2012). In rural educational environments, teachers and students may be more aware of individuals' unique needs and, as a result, develop protective factors for these students (Carlson & Matthes, 1987; Dunne, 1983). For instance, educators may be more prone to excuse, tolerate, or ignore socially awkward behaviors of a local student or his or her family due to the community value of supporting one another (Luse, 2017). This protective pattern continues in the community in the absence of severe injury or harm. However, these protective mechanisms do not support the necessary outcomes for future success in employment and quality of life, especially, if the student moves away to another community where he or she is not known (Smarick, 2017).

Defining Assistive Technology

According to the Individuals With Disabilities Education Act (2004), an assistive technology device refers to "Any item, piece of equipment or product system whether acquired commercially off the shelf, modified, or customized that is used to increase, maintain, or improve the functional capabilities of a child with a disability" (34 C.F.R. § 300.5). The most effective technologies are those that address a student's functional need, ability level, and student context differences (Camargo et al., 2014). The likelihood of general education teachers providing higher implementation fidelity increases when strategies involving assistive technologies are easy to implement, require minimal time for teacher-to-student assistance, and result in an immediate and noticeable change in student behavior (Fitzer et al., 2007; Morgan et al., 2016; Sutherland, Lewis-Palmer, Stichter, & Morgan, 2008). One type of assistive technology that has been demonstrated to possess these characteristics is video modeling (VM). VM has been identified by several researchers as a commonly used teaching tool, which involves a student viewing video footage of a model performing a targeted behavior or skill with the intention of the student later performing the targeted behavior or skill (Allen, Wallace, Renes, Bowen, & Burke, 2010; Buggey, 2009; Burton, Anderson, Prater, & Dyches, 2013; Cihak, Fahrenkrog, Ayres, & Smith, 2010). VM provides a concrete demonstration of what is required and how the behavior or skill should be performed.

Previous studies regarding the use of VM techniques for students with EBD incorporated the use of packaged programs, peers as models, or teacher-created videos with the student as the model (see Baker, Lang, & O'Reilly, 2009). Multiple researchers have described efforts taken by school personnel to create, use, and evaluate VM for students with more complex needs (e.g., autism spectrum disorder, physical disabilities; see Camargo et al., 2014). To date, there are no studies regarding the use of VM in which students with EBD select their own social behaviors for improvement,

create their own videos, and implement a VM strategy to improve their use of social skills.

Video Modeling as Assistive Technology

VM is not a new assistive technology. There are many programs and applications for the use of VM, which are low cost and portable, allowing easy use at school, home, and in the community (Carnahan, Basham, Christman, & Hollingshead, 2012; Nikopoulos, Canavan, & Nikopoulou-Smyth, 2009; Shane et al., 2012). Studies by several researchers indicate the use of mobile VM increases intrinsic and extrinsic motivation, as well as achievement, while also enhancing student engagement (Ciampa, 2014; Lin, Fulford, Ho, Iyoda, & Ackerman, 2012; West, 2012). The use of VM has been demonstrated as effective in teaching new skills across varying disabilities and settings for both academic and behavioral needs (see Baker et al., 2009; Blood, Johnson, Ridenour, Simmons, & Crouch, 2011; Burton et al., 2013; Clare, Jenson, Kehle, & Bray, 2000; Cumming, 2010; Hitchcock, Dowrick, & Prater, 2003; O'Reilly et al., 2005). The use of video-based technology can be highly motivating for many students and is a useful tool for providing basic instruction that includes the modeling of appropriate behavior and the provision of feedback (Bellini & Akullian, 2007; Sherer et al., 2001; Sturmey, 2003).

Types of VM

Shukla-Mehta, Miller, and Callahan (2010) identified four types of VM that have typically been used to teach a range of skills across settings including (a) with other as a model (VMO), (b) point of view video modeling (POV), (c) video prompting (VP), and (d) video self-modeling (VSM). VMO involves recording someone performing the desired behavior other than the observer student. For example, a student may watch a video of a peer performing an appropriate greeting. When using POV, the desired behavior or skill is recorded from the perspective of the observer student. For example, a student observing a model of an appropriate handshake would view a clip of another person approaching, extending their hand, and then a hand extending from the perspective of the viewer to meet that other persons' hand. VP involves creating a task analysis of the desired behavior or skill and recording each step with an added pause for the observer student to attempt the step before viewing subsequent steps. VP can include the observer student or someone other than the observer student modeling the desired behavior or skill. Characteristics of the most effective models include individuals who (a) are close in age to the observer, (b) are of the same gender and race, (c) possess similar personality traits, (d) tend to perform the skills modeled at a level slightly higher than the observer, (e) are admired by the observer for demonstrating the behavior, and (f) are rewarded

for demonstrating the desired behavior or skill with a reward desired by the observer model (Bandura, 1997; Thoresen & Hosford, 1973). VSM differs from other types of VM, in that, the observer student acts as the model for the behavior or skill performance, and the video is used for later viewing by that student. Considering that students with EBD characteristically have difficulty maintaining relationships with peers and adults (Cook et al., 2008), having them act as their own models in VSM for appropriate social interaction skills is compelling.

Focus on VSM Use

VSM is an antecedent-based intervention that involves students observing themselves modeling the accurate performance of a target behavior at a more advanced level than they currently do and then reproducing the behavior in identified situations or settings (Dowrick, 1999; Kern & Clemens, 2007). In VSM, undesired or unsuccessful behaviors are omitted through editing, and the student watches himself or herself effectively performing the desired behavior (Buggey, 2007). VSM has been demonstrated to be an effective evidence-based intervention for improving a range of functional skills (Hitchcock et al., 2003) and has been shown to produce behavior change that persists over time (Bellini & Akullian, 2007).

One way to use VSM is to embed it within an electronic or digital book (i.e., e-books). E-books are a popular and widely available assistive technology application, which allows students to create short interactive books to digitally tell their own story. Students can embed selected audio, text bubbles, and/or pictures and graphics within the video e-books. E-books and VSM can be used to promote students' independent skill use through personalization, ease of mobility, and the provision of rich detail about where, when, and with whom identified social skills can be used. Using the e-book as a vehicle for students to create and implement VSM alleviates the teachers' need to do so and relieves difficulties cited in previous studies on teacher-implemented VSM (Fitzer et al., 2007; Morgan et al., 2016; Sutherland et al., 2008).

Case Examples of the Use of VSM With Students With EBD

The following describes the successful use of VSM using an e-book application as an assistive technology to generate the use of social skills. The three ninth-grade students with EBD, Marcus, Claire, and Darren, were attending their first year at a large rural high school that served 690 total students in Grades 9–12. This rural high school produced a graduation rate of 87%, which was higher than the state average of 83%. Notably, it had a higher student to teacher ratio of 21:1 than the state average of 19:1.

Decision to Use VSM

Marcus, Claire, and Darren received special education services in the high school and had been taught social skills using several packaged programs during their years in middle school. Social skill evaluations (e.g., Social Skills Improvement System–Rating Scales [SSIS]; Gresham & Elliott, 2008) indicated that they possessed critical prerequisite skills to social interactions (e.g., communication) and could verbally repeat the steps to perform age-appropriate social skills; however, they had performance deficits in using the skills. Data indicated that traditional social skills instruction at response to intervention Tiers 1 and 2 levels had been ineffective for all three students and that they were currently failing in their academic classroom settings. As reported by the general education teachers, the students' failing grades were a result of a lack of social competence.

A multidisciplinary team decided that these students needed a strong personalized intervention that focused on the instruction of a replacement behavior that would provide each student with functionally relevant reinforcers to ensure independence with minimal adult interference. The team agreed that VSM might fit the needs of the students as each could act as his or her model and bypass the difficulty of finding another individual who met the characteristics listed for effective models.

The school computer laboratory included the interactive e-book creator program for iPad (i.e., Red Jumper Limited). The team members agreed that using this e-book application would allow the students to create interactive, personalized self-modeling videos, and that the use of this assistive technology to teach functionally equivalent social skills would fit as a Tier 3 support (Maag, 2005; Mitchell, Stormont, & Gage, 2011). Subsequently, the team convened a meeting with each student, their caregivers, and teachers. At the meeting, everyone was provided with data from the SSIS. The students were then asked to use the data to identify the social skill(s) they viewed as necessary for their success in one inclusive class of their choice. The teachers then described the VSM procedures for creating e-books and asked the students whether they would be willing to try the technique. All three students agreed to participate.

The teachers then scheduled times for the students to attend the school's computer laboratory to design and develop their VSM e-books. One of the science teachers volunteered to provide technical support for the students while they were in the computer laboratory and helped the students by providing directions on how to use the e-book application.

Development and Implementation Procedure of the VSM E-Book

Once the students learned to use the e-book application, the science teacher directed the students to construct a list of

Table 1. Marcus' Steps for Joining a Conversation.

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- | |
|--|
| 1. Listen for the topic being discussed |
| 2. Think about what you know about the topic |
| 3. Decide what you want to add—use a note card to write your thoughts if needed |
| 4. Choose the time/way to join in—whether this is a raise the hand or jump in discussion |
| 5. Join in the discussion |
-

steps required to perform their identified social skills for use on a storyboard sequence. The teacher provided the book *Skillstreaming the Adolescent: A Guide for Teaching Prosocial Skills* (McGinnis, 2016) to guide them. This step-by-step social skills format was familiar to the students as they had previously been taught using similarly formatted programs in middle school. The steps, adapted by Marcus, for joining in a discussion are outlined in Table 1. After writing out the skill steps, each student, with the help of the volunteer science teacher, created a video sequence on a storyboard of their VSM e-book. See Figure 1 for example by Marcus of his initial storyboard for Social Skill 10, joining a class discussion.

Next, each student identified a method for monitoring and then reporting his or her progress to his or her selected general education teacher at brief weekly meetings. Although the teachers agreed to encourage the students to use the VSM assistive technology e-books and acted as support personnel to ensure progress, the strategy was primarily implemented by the three students.

Each student engaged in a multistep process for creating and using the e-book format for VSM. First, each student made videos of himself or herself demonstrating the selected social skill steps, individually or interacting with a teacher and/or other students in the general education class environment. The students recorded the model within the environment in which they identified a need for improvement in the use of social skills. Videos were edited to remove errors, prompts, and inappropriate behavior. After collecting video examples of the social skills the students deemed to be performed at their expertise level, they designed the layout of their e-book using a storyboard. While designing her e-book, Claire used advanced components of the e-book application such as the pen tool to add drawings, arrows, or other graphics for emphasis, text prompting, and reinforcement. In addition, she added her own narrations and applied a range of layouts. Although Marcus and Darren used fewer advanced design features, they gradually increased the sophistication of their books as they became comfortable with the application. After completion of the storyboards and when the students felt they had acquired the necessary video, audio, and graphics, they composed the e-book. Their completed e-books range from 3 to 6 min in duration.

After completing their e-books, the students began to implement their use for skill development. Students were

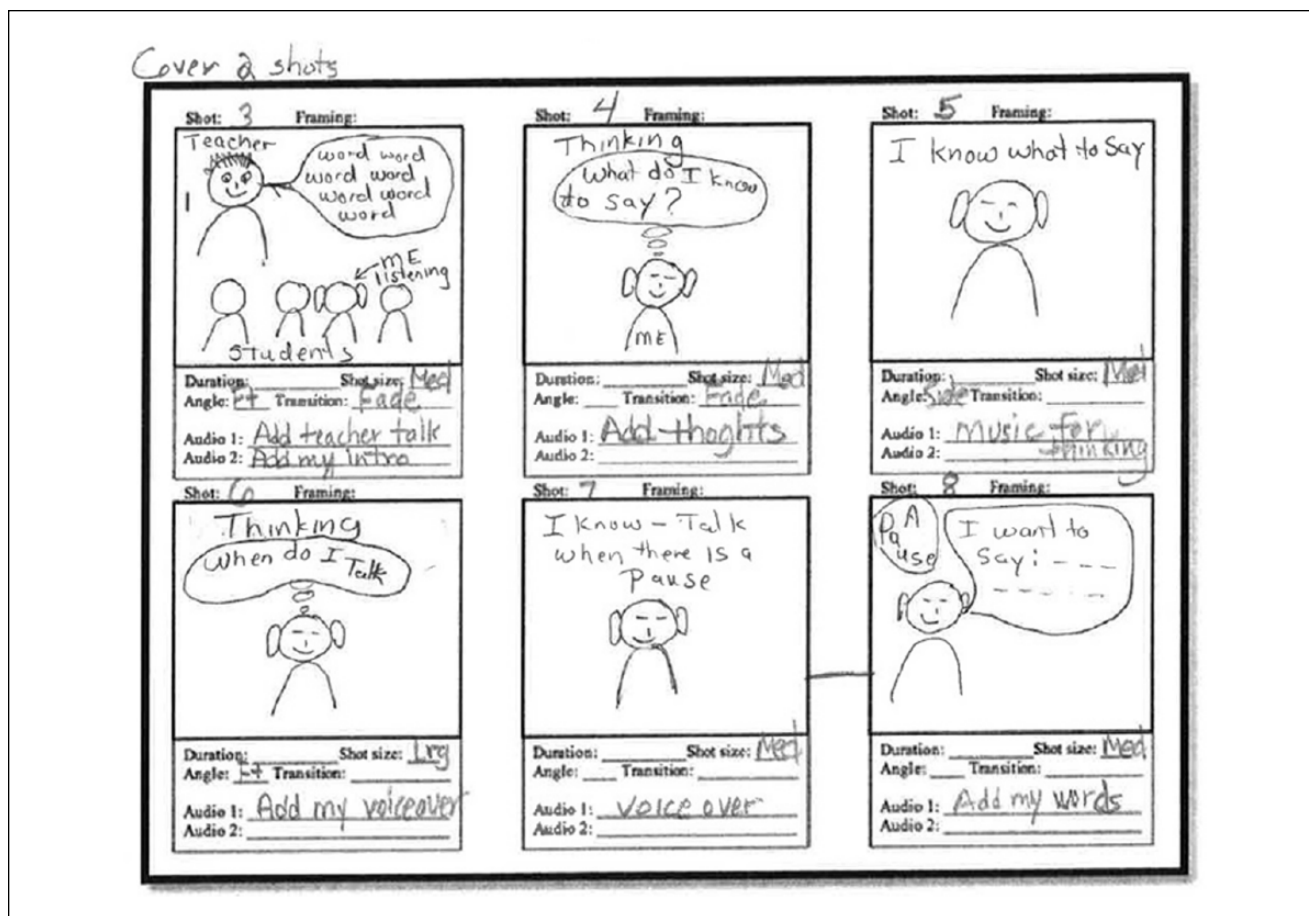


Figure 1. Example by Marcus of initial storyboard for joining a class discussion.

directed to watch their e-books and were given opportunities to show their videos to their teacher and peers across several days while in the special education setting. Students often watched the e-book between 3 and 5 times a day prior to performance attempts. The students continued to watch their e-book and rehearsed the desired behavior until they reported they were confident to perform the skill in the general education setting. At that time, they each let their general education teacher know they were ready. The students watched their e-book just before going into the general education classroom, performed the skill during the class session, and discussed their performance during a weekly session with the general education teacher. During these sessions, the students reflected on their performance, needed to adapt the demonstration of the skill within their e-book and try again, or to adapt the demonstration of the skill within their e-book.

Collection of Social Validity Data

Social validity was assessed using an open-ended questionnaire created by a behavior consultant (first author) to solicit the participants' opinions concerning the significance

of their intervention goals, the acceptability of the procedures, and their evaluation of the importance of the results (Gresham & Lopez, 1996). Researchers found that when interventions were considered acceptable, they were more likely to be implemented (Gresham, MacMillan, Beebe-Frankenberger, & Bocian, 2000; Lane et al., 2009). The social validity questionnaires given to the general education teachers, the paraeducators, the students' caregivers, and the students were composed of Likert-type and open-ended questions asking about their perceptions of using the assistive technology, the influence on the students' behavior, and suggestions for further use.

Results

Teacher, Paraeducator, and Parent Social Validity Data

The questionnaire for the teachers and paraeducators consisted of six questions including four open-ended questions and two Likert-type scale questions. Open-ended questions were as follows: (a) What specific changes or influences, if any, did you observe in the student's behavior(s) that you

attribute to the social skill VSM e-book? (b) Describe how you think the VSM e-book influenced student behavior in class (at home). (c) Do you have suggestions for the future use of the social skill VSM e-book? and (d) Please share any other thoughts you have with regard to the social skill VSM e-book. The two Likert-type scale questions share the same scale (i.e., 1 = *not at all*, 2 = *a little bit*, 3 = *same as previous*, 4 = *a bit more than previous*, and 5 = *very engaged*) and were as follows: (a) To what degree did you think your student participated in social skill development because of using VSM e-book? and (b) What level of participation did your student show in class (at home) for demonstrating self-identified social skill(s)?

The classroom teachers and paraeducators reported that they observed an immediate change in the students' attitude (e.g., implementing steps for Social Skill 15, dealing with a disagreeable situation), attempts at interactions with peers and adults (i.e., used steps for Social Skill 10, joining and participating in a discussion), and application of social skill use to complete assignments (i.e., Social Skill 25, negotiating with a teacher for modification of due date or assignment product outcome). One teacher remarked, "I did not realize how smart these students were until I saw them using the iPad as a tool for learning with ease and enthusiasm." One of the two paraeducators commented, "I couldn't believe the change in the students—they participated more in class discussions with relevant contributions better than some of the typical students," and the other paraeducator wrote, "Amazing, I can't wait to try this technique with some other students." The technology volunteer stated,

I use technology a great deal in my science class, for yearbook sponsorship, and in my elective media class. After seeing these sullen students who barely moved their bodies in the hallway transformed into vibrant students through this project, I can't wait to explore other assistive technologies to use at our school.

Parent responses to the questionnaire suggested they perceived their children benefited from the use of the assistive technology. Darren's mother wrote, "I have never seen my son want to do homework and talk about new vocabulary he was learning in class. Unbelievable!" Claire's father wrote, "Claire loves her iPad program and has even suggested that I may benefit from making a VSM e-book for myself. It is the most we have interacted for several years. Thank you." Marcus's parent wrote, "This program has helped him come out of his shell. He has even used the negotiation skill a couple of times at home and accepted no in a very kind way. We were shocked." All the caregivers indicated that they had positive experiences and saw changes that surprised them about their child's social competence outside school and their engagement with schoolwork.

Student Social Validity Data

The questionnaire for the students also consisted of six questions: four open-ended questions and two Likert-type scale questions. Open-ended questions were as follows: (a) What specific changes or influences, if any, did you observe in your social behavior(s) that you attribute to the social skill VSM e-book? (b) Describe how you think the use of the VSM e-book influenced your behavior in class (at home). (c) Do you have suggestions for the future use of the social skill VSM e-book? and (d) Please share any other thoughts you have with regard to the social skill VSM e-book. The two Likert-type scale questions share the same scale (i.e., 1 = *not at all*, 2 = *a little bit*, 3 = *same as previous*, 4 = *a bit more than previous*, and 5 = *very engaged*) and were as follows: (a) To what degree did you think you participated in the social skill development because the VSM e-book? and (b) What level of participation did you show in class (at home) for demonstrating your identified social skill(s)?

The three students wrote comments on their social validity questionnaire that indicated they liked having control over making their VSM e-book and using it. Darren wrote, "I liked the program and the way I could say what I wanted to learn and what skill I wanted to use." Marcus reported, "I loved doing this, and it made sense in class. I liked deciding how and when I wanted to work on talking in class." Claire indicated,

This was my work, not something that someone told me to do. I had fun at school instead of listening to someone in my face all the time about how I could do the work if I just tried. Not boring at all.

All three students participated and carried their iPads with them every day. The only tentative comment came from Marcus. He wrote, "I wasn't sure I could trust this, so it wouldn't be used against me later. It wasn't; so it turned out good."

Another indication of whether a strategy possesses social validity is reflected in its sustained use by relative stakeholders. The students continued to use the e-books across other settings and began to include other individuals in their videos. Claire created an additional e-book relating to the skill of negotiating. She included in her e-book a video depicting her negotiating with her teacher for extra time on an assignment. See the cover example showing the audio button created by Claire in Figure 2. She shared this e-book with students in the elementary-level special education program. Claire reported that the VSM strategy allowed her to demonstrate her own approach to asking for help. Her e-book included her video and audio recording of talking with her teacher that accompanied graphics and pictures of the steps for the social skill.

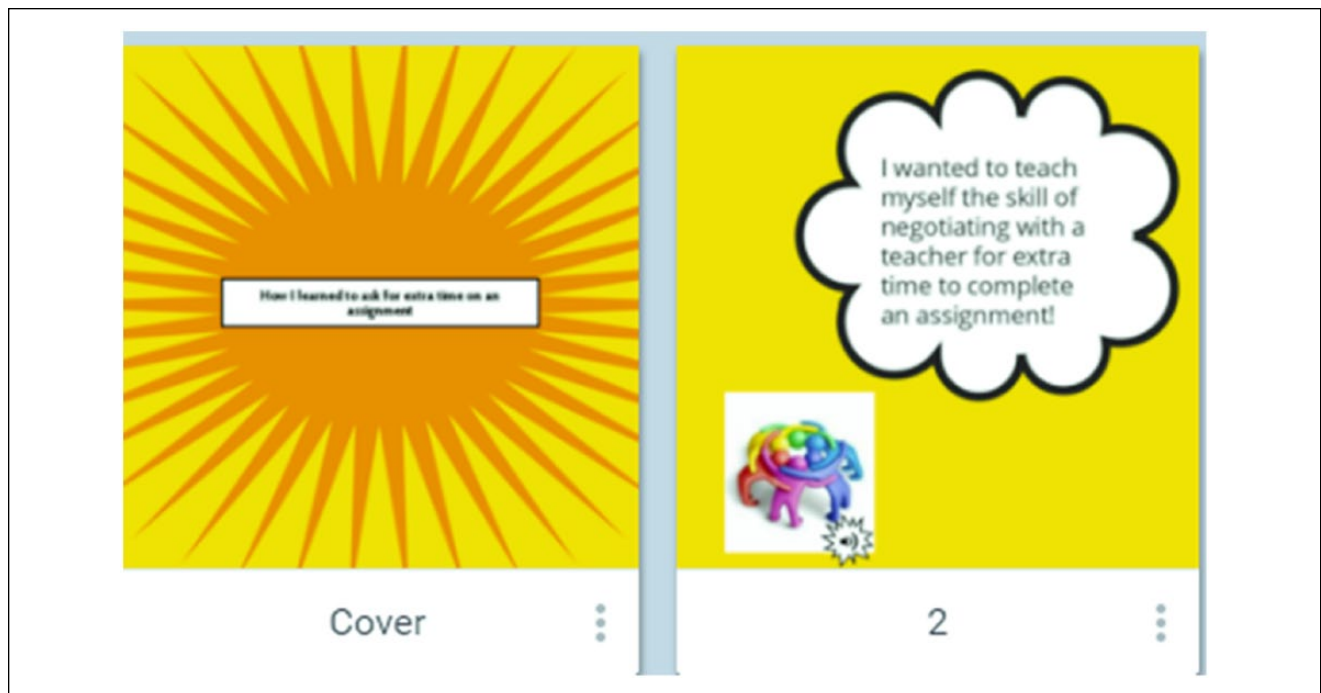


Figure 2. Example of e-book cover with audio button created by Claire.

Conclusion

The use of VSM has great potential for students with EBD in the elimination of barriers to sustaining and maintaining relationships. In the case example described above, we provided a demonstration of the use of VSM and e-book assistive technology to promote students' use of social skills in the general education setting. Furthermore, we showed that the intervention package was perceived as enjoyable by the participants, which is critical in light of data suggesting these students often fail to participate in social skills instruction (Gresham, 2002). Finally, the package was deemed to be effective by the students' teachers and parents.

Teachers in rural settings face unique challenges in providing quality education (Berry, Petrin, Gravelle, & Farmer, 2011) to students with EBD, including decreased fiscal resources, opportunities for high-quality professional development, Internet access, and increased geographical isolation (Berry et al., 2011; Howley, Wood, & Hough, 2011; Whitacre & Mills, 2010). The use of VSM e-books may help teachers circumvent some of these challenges, in that, they can be produced with inexpensive technology and limited Internet access. Furthermore, they require minimal training, are portable, and are student directed, requiring minimal instructional time by a teacher (Burton et al., 2013; Cihak et al., 2010).

The incorporation of technology in social skills instruction appeared to facilitate students' engagement during

role-play and e-book planning. Interestingly, the students practiced appropriate social skills during the development of the e-books and expressed enjoyment in practicing with their peers prior to implementation into general education setting. Furthermore, it may have promoted students' accountability during implementation of the strategy. Researchers have suggested that the feeling of ownership may result in the use of targeted skills in other contexts (Maag, 2005). All three participants invited their friends to join them in creating VSM e-books, formed a school club, and developed a Google Docs page dedicated to VSM e-books created by many students in the school.

The current case study highlights the potential of VSM e-books for use with this unique population of students, but there is a need for further investigation to demonstrate causal relations between the intervention and changes in student behavior. Researchers also should assess the effects of student selection of social skill targets on the generalized use of those targets in novel contexts. Finally, it is critical that researchers evaluate the role of learner characteristics in the effectiveness of this self-directed package including students' ability to identify social skills in need of improvement, preference for the use of technology, and motivation toward interacting with others.

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